



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam: : Trimester: Fall - 2024

Course Code: CSE 4889/CSE 489, Course Title: Machine Learning

Total Marks: 30

Duration: 90 minutes

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Please answer all the questions. Figures in the right-hand margin indicate full marks.

Question 1:

5+3+2=10

- A. Explain the Machine Learning life-cycle process.
- B. What do you understand by 10-Fold Cross Validation? Find the Accuracy, Precision and F1-Score from **Table1**.

Table 1

TP = 480	FN = 120
FP = 50	TN = 350

- C. Suppose, you want to find out a way to handle the test center crisis for the Dengue outbreak in our country. Therefore, you have built a system with a satisfactory accuracy score to check if the person is dengue positive or negative based on some specific input. However, when you used the system in practice, you noticed that the number of patients increased drastically in hospitals and it became too rushed. You observed that the system sometimes shows positive results even for the people with general seasonal fever. In order to identify the problems, explain the importance of **Specificity** and **Sensitivity** here.

Question 2:

5+5=10

- A. The yield y of a chemical process is a random variable whose value is considered to be a linear function of the temperature x . The following data of corresponding values of x and y are:

Table 2

Temperature in Celsius (x)	0	25	50	75	100
Yield in Grams (y)	14	38	54	76	95

The average and standard deviation of temperature and yield are:

Avg. $x = 50$, $s_x = 39.628$, Avg. $y = 55.4$, $s_y = 31.667$

Now, generate the Linear Regression model for the above problem.

- B. Explain the need of using One-vs-all method with proper explanation. State the pros and cons of using a Logistic Regression classifier.

Question 3:

5+5=10

- A. Find the Root node from Table 3 using CART algorithm

Table 3

Instance	X1	X2	Classification
1	T	T	Y
2	T	T	Y
3	T	F	N
4	F	F	Y
5	F	T	N
6	F	T	N

- B. Explain Random Forest and Bagging techniques with their advantages and disadvantages.